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The University of Toronto Climate Downscaling Workflow: Tools and Resources for Climate Change Impact Analysis

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Software

- Review
- Repository
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Summary

The University of Toronto Climate Downscaling Workflow (UTCDW) is a resource designed to guide users through the process of producing their own statistically downscaled climate projections for the purpose of climate change impact analysis. The target audience includes graduate students or practitioners in engineering, the physical, mathematical, and computational sciences. Such individuals may work outside the field of climate science, but are interested in how their subject of study is sensitive to climate change. The main component of the UTCDW is a Jupyter Book we call the "UTCDW Guidebook". The Guidebook introduces users to basic climate science concepts and beginning-to-intermediate concepts in the application of statistical climate downscaling. It also works through the decisions that must be made when designing a climate change impact study. Finally, it demonstrates how to download climate data, do exploratory analysis and model validation, and provides examples of end-to-end workflows for small climate downscaling projects. The Guidebook was written for self-study by members of the target audience, though it may be included as a part of a course on climate change impact analysis for graduate students or upper-year undergraduate students in the applied sciences. All of the Guidebook source materials (Jupyter Notebooks, markdown files, and Python scripts) are accessible in the UTCDW_Guidebook GitHub repository.

Statement of Need

Important natural and human-designed systems studied under disciplines like ecology, engineering, and public health are sensitive to weather and climate conditions. This means these systems are also sensitive to the impacts of anthropogenic global warming and climate change. As climate change intensifies, these systems may become more vulnerable to climate risk. Therefore experts in these fields need to assess climate change risks for the purposes of adaptation. However, unprocessed climate model output such as those from the Coupled Model Intercomparison Project Phase 6 (CMIP6) (Eyring et al., 2016) archive usually do not directly meet their requirements. Models are skillful but inherently imperfect, so their output is typically biased with respect to observed local climates. In addition, raw climate model variables do not directly correspond to useful metrics for calculating loads and hazards are are often not available on the required spatial or temporal scales. *Downscaling* refers to a set of procedures that adjust and map climate data to variables, times, and locations, relevant for discipline-specific applications. It involves physical reasoning, calibration, and statistical bias-correction of climate model output using observations and analysis requirements for specific study domains. Downscaling is enriched with an understanding of the climate system and

subject matter expertise. Engineering firms, government departments, and non-profit organizations carry out downscaling routinely to help various clients and stakeholders adapt to a changing climate and seek effective solutions to the dangerous effects of climate change.

Unfortunately, there is no single accepted standard for downscaling, only a range of methods that are centred on the practice of different climate-service providers. This makes it hard for new researchers to get started and perform their own analysis, especially when existing downscaled data products do not suit their needs. The UTCDW documents state-of-the-art methods for statistical bias correction and downscaling that are relatively straightforward for users to implement, and helps users build technical competence in the early Guidebook chapters. Other educational materials on climate change impact assessment are available, but they are either limited regarding the complexity of the methods covered (Anderson & Smith, 2021), lack code examples to help new users actually work with the data they need (Kotamarthi et al., 2021), or are too advanced for a first introduction to downscaling (Maraun & Widmann, 2018). The UTCDW Guidebook fills this gap by providing a basic introduction to concepts in climate science and including worked examples and code for downscaling methods appropriate for applications. This makes the UTCDW ideal for users who are new to these concepts and wish to be able to start doing data analysis for their impact study as quickly as possible.

Guidebook Content

The UTCDW Guidebook consists of six chapters. The first four chapters introduce the reader the background knowledge required to perform downscaling. Chapter 1 discusses the purpose of the book, points to some prerequisite material, and explains how to set up the Python computing environment. Chapter 2 is an introduction to climate science and climate modeling, so readers build an understanding of how climate change projections are made and what their limitations are. Chapter 3 begins to demonstrate work with real climate data. It shows users how to access observational, reanalysis, and climate model data to use for an impact study, and how to do exploratory analysis with observational and raw climate model data. Chapter 4 contains explanations of various methods of bias-correction and downscaling, and examples of how to validate historical downscaled model output and assess future projections. The recommended bias-correction methods are the quantile-mapping based methods proposed by Cannon et al. (2015) and implemented in the `xclim` package (Logan et al., 2023), and the recommended downscaling methods, which enhance the spatial resolution of gridded data, are the constructed-analogue (CA) based methods described in Werner & Cannon (2016). Python code for the CA methods is provided in the GitHub repository, leveraging the utility of `xclim` and the Pangeo software ecosystem.

The next two chapters focus on applying the knowledge and methods covered in Chapters 1-4. Chapter 5 introduces the "Downscaling Workflow" part of the UTCDW. This chapter unpacks the decisions that must be made when designing a climate change impact study and breaks the analysis tasks down into digestible steps. Chapter 6 contains several simple examples of the workflow. The examples implement both bias-adjustment of one-dimensional time series for a point location and spatial resolution-enhancing downscaling methods for a larger geographical area. Their purpose is to demonstrate the end-to-end process of stating a problem, acquiring the necessary data, producing calibrated projections of metrics relevant to the application at hand, and quantifying uncertainty in the projections. For users who wish to use pre-processed downscaled data, the Guidebook also shows users how to access ready-to-use downscaled model output for Canada from the Pacific Climate Impacts Consortium (Consortium, 2021). The Guidebook focuses on Canadian regions and observational data products, though we include links to available observational and downscaled data products for additional regions as well as global data.

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The UTCDW also includes a website that contains a guided survey that helps a user design their climate impact study and generates a flowchart image that lays out the steps for the workflow. While this website is separate from the Guidebook, the flowcharts are featured in the worked examples (Chapter 6) to explain the procedure before showing the steps of the analysis. We welcome user submission of additional worked examples to contribute to the gallery of examples to be included on the website, or future versions of the UTCDW Guidebook

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